

## Task-3

### PLSQL Procedure for Loops

Aim-  
To write PLSQL programs using loops for printing prime numbers customer IDs and for demonstrating loop control in different scenarios.

#### PROCEDURE

- Start a PLSQL block or procedure.
- Use a cursor (if required) to fetch customer IDs from table.
- For each ID, check whether it is a prime number using a loop.
- Use FOR LOOP/WHILE LOOP to demonstrate prime number checking.
- Print the result using DBMS\_OUTPUT.PUT\_LINE.
- End the block.

Example 1: Using WHILE LOOP with cursor  
prime check using WHILE loop

```
CREATE OR REPLACE PROCEDURE print_prime_customers IS
    CURSOR cust_cur IS
        SELECT customer_id FROM customers;
    v_id NUMBER;
    v_is_prime BOOLEAN;
    v_i NUMBER;
BEGIN
    OPEN cust_cur;
    LOOP
        FETCH cust_cur INTO v_id;
        EXIT WHEN cust_cur%NOTFOUND;
        IF v_id < 2 THEN
            v_is_prime := FALSE;
        ELSE
            v_is_prime := TRUE;
            v_i := 2;
            WHILE v_i <= TRUNC(SQRT(v_id)) LOOP
```



```

IF MOD (v-id, v-i) = 0 THEN
    v-is-prime := FALSE;
    EXIT;
END IF;
v-i := v-i + 1;
END LOOP;
END IF;
IF v-is-prime THEN
    DBMS_OUTPUT.PUT_LINE('prime customer ID: ' || v-id);
END IF;
END LOOP;
CLOSE 'cust-cur';
END;

```

This procedure checks all customer IDs in the table and prints the prime ones using a while loop.

Example 2:- Using FOR loop for first N prime numbers

CREATE OR REPLACE PROCEDURE print-first-n-primes(n NUMBER) IS

```

    v-num NUMBER := 2;
    v-count NUMBER := 0;
    v-is-prime BOOLEAN;

```

BEGIN

```

    WHILE v-count < n LOOP

```

```

        v-is-prime := TRUE;

```

```

        -- prime check using for loop

```

```

        FOR i IN 2..TRUNC(SQRT(v-num)) LOOP

```

```

            IF MOD(v-num, i) = 0 THEN

```

```

                v-is-prime := FALSE;

```

```

                EXIT;

```

```

            END IF;

```

```

        END LOOP;

```

```

        IF v-is-prime THEN

```

```

            DBMS_OUTPUT.PUT_LINE('prime: ' || v-num);

```

```

            v-count := v-count + 1;

```

```

        END IF;

```



$V\_num := V\_num + 1$ ;

END LOOP;

END;

This procedure prints the first  $n$  prime numbers using a  $for$  loop

As example:

BEGIN

  print-First-n-primes ( $n$ ); -- prints first  $n$  prime numbers

END;

RESULT:- Thus, the write out side program's using loops ~~As~~ printing prime number customer IDs and ~~As~~ demonstrating loop control in different scenarios is done successfully



## TASK:- PL/SQL Procedure & Loops

SQL\*Plus: Release 11.2.0.2.0 Production on Wed Oct 15 21:34:44 2025

Copyright (c) 1982, 2014, Oracle. All rights reserved.

SQL> connect

Enter user-name: system

Enter password:

Connected.

SQL> CREATE OR REPLACE PROCEDURE print\_prime\_customers IS

```
2  CURSOR cust_cur IS
3  SELECT customer_id FROM customers;
4
5  v_id    NUMBER;
6  v_is_prime BOOLEAN;
7  v_i     NUMBER;
8 BEGIN
9  OPEN cust_cur;
10 LOOP
11  FETCH cust_cur INTO v_id;
12  EXIT WHEN cust_cur%NOTFOUND;
13
14  -- Prime check
15  IF v_id < 2 THEN
16  v_is_prime := FALSE;
17  ELSE
```

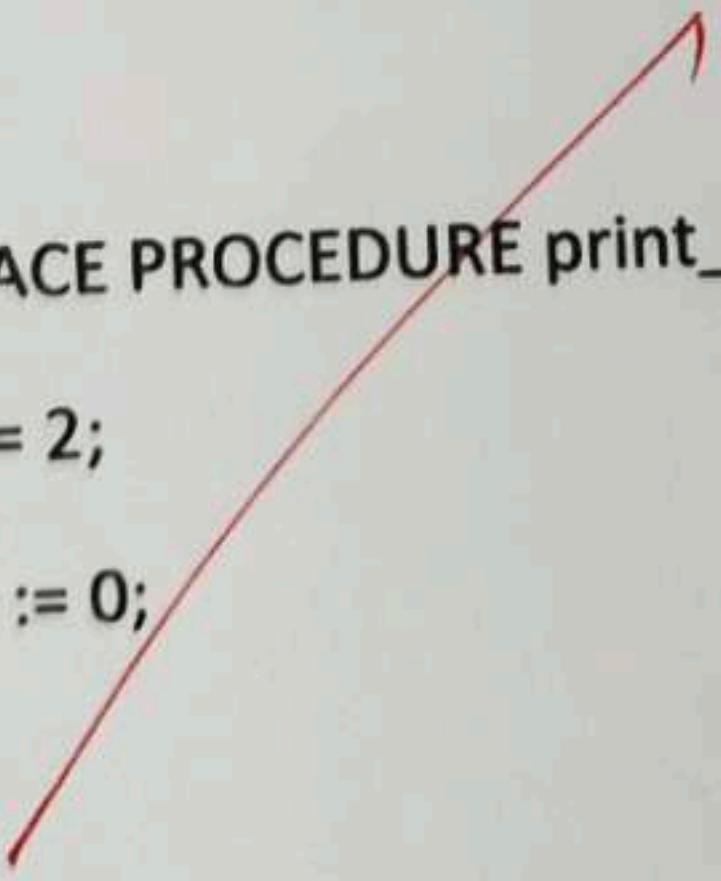


```
18  v_is_prime := TRUE;
19  v_i := 2;
20
21  WHILE v_i <= FLOOR(SQRT(v_id)) LOOP
22    IF MOD(v_id, v_i) = 0 THEN
23      v_is_prime := FALSE;
24      EXIT;
25    END IF;
26    v_i := v_i + 1;
27  END LOOP;
28  END IF;
29
30  IF v_is_prime THEN
31    DBMS_OUTPUT.PUT_LINE('Prime customer ID: ' || v_id);
32  END IF;
33
34  END LOOP;
35  CLOSE cust_cur;
36  END;
37 /
```

Warning: Procedure created with compilation errors.

SQL> CREATE OR REPLACE PROCEDURE print\_first\_n\_primes(n NUMBER) IS

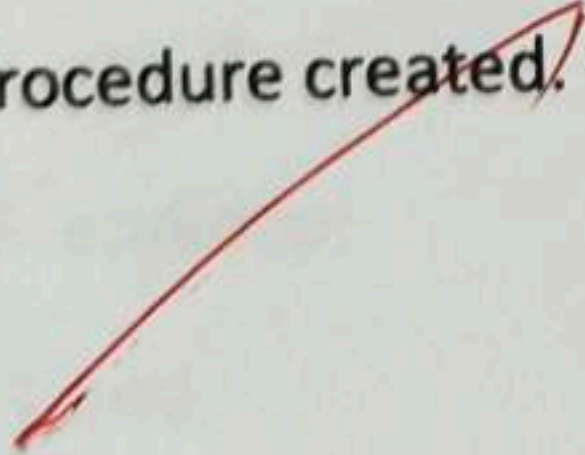
```
2  v_numNUMBER := 2;
3  v_countNUMBER := 0;
```





```
4  v_is_prime BOOLEAN;
5  BEGIN
6  WHILE v_count < n LOOP
7    v_is_prime := TRUE;
8
9    -- Prime check using FOR loop
10   FOR i IN 2 .. TRUNC(SQRT(v_num)) LOOP
11     IF MOD(v_num, i) = 0 THEN
12       v_is_prime := FALSE;
13       EXIT;
14     END IF;
15   END LOOP;
16
17   IF v_is_prime THEN
18     DBMS_OUTPUT.PUT_LINE('Prime: ' || v_num);
19     v_count := v_count + 1;
20   END IF;
21
22   v_num := v_num + 1;
23 END LOOP;
24 END;
25 /
```

Procedure created.





```
SQL> declare
2 lonumber(3);
3 hinumber(3);
4 nnumber(2);
5 mnumber(2);
6 cnumber(20);
7 begin
8 dbms_output.put_line('enter the customer id from to limit:');
9 lo:=&lo;
10 hi:=&hi;
11 for n in lo.. hi
12 loop
13 c:=0;
14 for m in 1..n
15 loop
16 if mod(n,m)=0 then
17 c:=c+1;
18 end if;
19 endloop;
20 if c<=2 then
21 dbms_output.put_line(n || '\n');
22 end if;
23 endloop;
24 end;
```



25 /

Enter value for lo: 101

old 9: lo:=&lo;

new 9: lo:=101;

Enter value for hi: 120

old 10: hi:=&hi;

new 10: hi:=120;

PL/SQL procedure successfully completed.

SQL> declare

2 bknumber(5);

3 snumber:=0;

4 r number;

5 len number;

6 m number;

7 begin

8 bk:=&bk;

9 m:=bk;

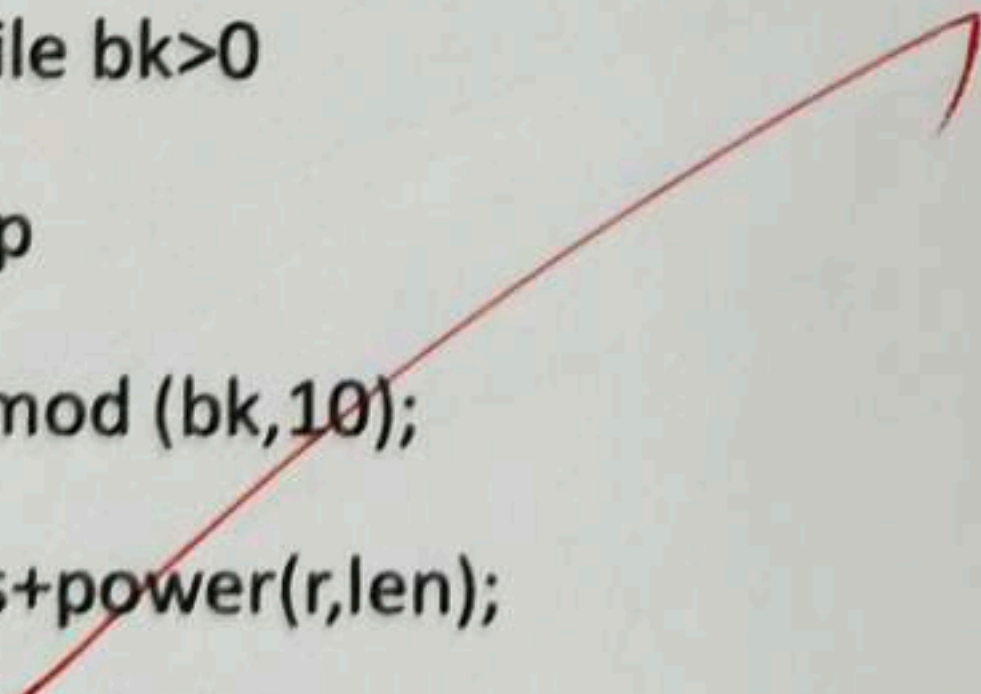
10 len:=length(to\_char(bk));

11 while bk>0

12 loop

13 r:=mod (bk,10);

14 s:=s+power(r,len);





```

15 bk:=trunc(bk/10);
16 endloop;
17 if
18 m=s
19 then
20 dbms_output.put_line('given number is armstrong');
21 else
22 dbms_output.put_line('given number is not an armstrong');
23 end if;
24 end;
25 /

```

Enter value for bk: 234

old 8: bk:=&bk;

new 8: bk:=234;

PL/SQL procedure successfully completed.

SQL>

| VEL TECH                |    |
|-------------------------|----|
| EX NO.                  | 2  |
| PERFORMANCE (5)         | 5  |
| RESULT AND ANALYSIS (5) | 5  |
| VIVA VOCE (5)           | 4  |
| RECORD (5)              | 5  |
| TOTAL (20)              | 18 |
| SIGN WITH DATE          |    |

u/10/25